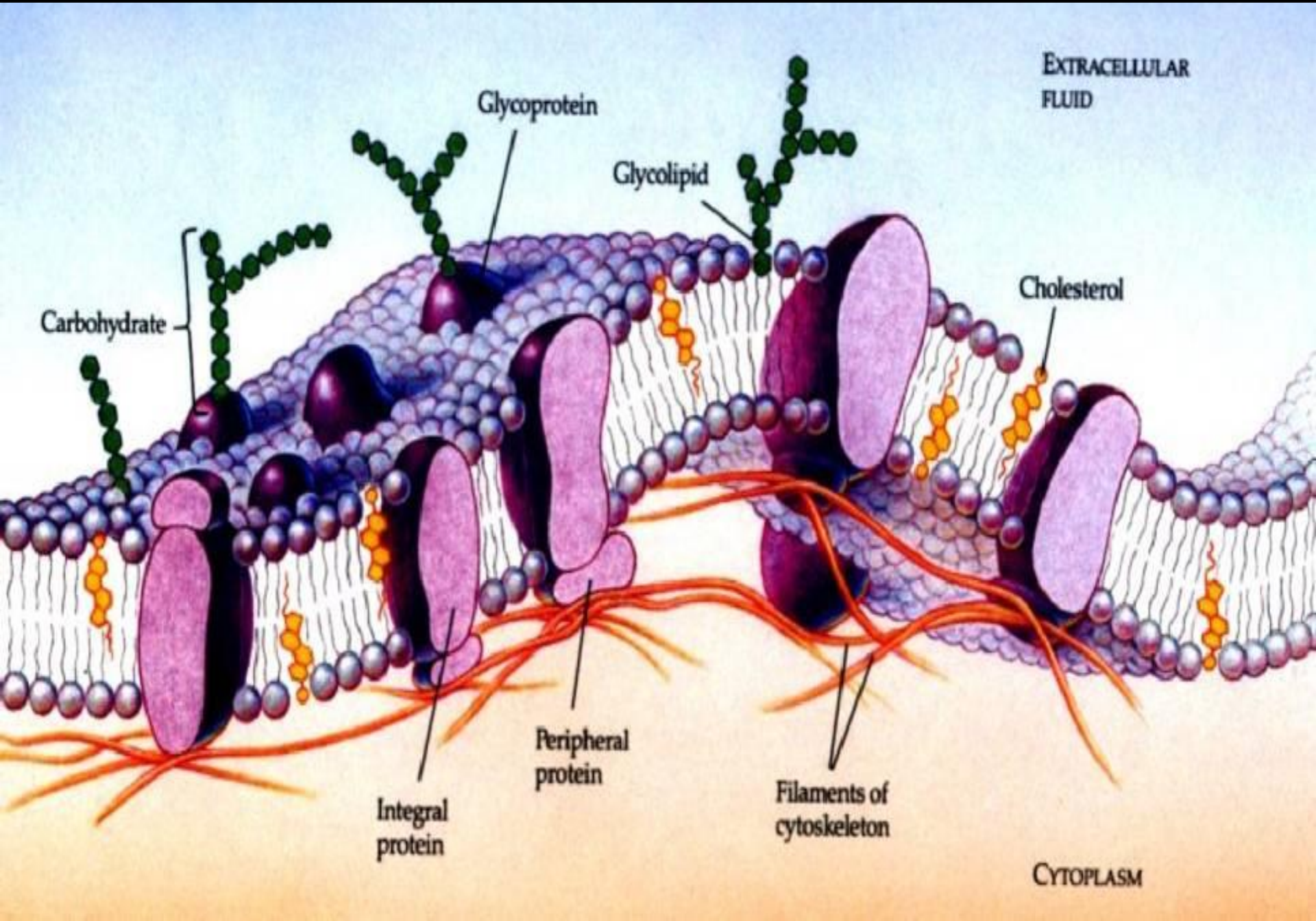
28/09/2026	Introduction	Monday	ARTS-LR7
02/10/2026	Chemistry and Molecules	Friday	ARTS-LR6
05/10/2026	Proteins	Monday	ARTS-LR7
09/10/2026	DNA	Friday	ARTS-LR6
12/10/2026	Cell membranes I	Monday	ARTS-LR7
16/10/2026	**Cell membranes II**	**Friday**	**ARTS-LR6**
19/10/2026	Energy	Monday	ARTS-LR7
23/10/2026	SPARE	Friday	ARTS-LR6
26/10/2026	The cytoskeleton	Monday	ARTS-LR7
30/10/2026	Biophysical techniques I	Friday	ARTS-LR6
02/11/2026	Biophysical techniques II	Monday	ARTS-LR7
06/11/2026	Diffusion	Friday	ARTS-LR6
09/11/2026	Thermodynamics of Reactions I	Monday	ARTS-LR7
13/11/2026	Thermodynamics of Reactions II	Friday	ARTS-LR6
16/11/2026	Blood	Monday	ARTS-LR7
20/11/2026	Biophysics of Nerve Cells I	Friday	ARTS-LR6
23/11/2026	Biophysics of Nerve Cells II	Monday	ARTS-LR7
27/11/2026	Biophysics of Nerve Cells III	Friday	ARTS-LR6
30/11/2026	Biophysics in Health and Disease	Monday	ARTS-LR7
04/12/2026	Careers in Biophysics	Friday	ARTS-LR6
07/12/2026	SPARE	Monday	ARTS-LR7
11/12/2026	SPARE	Friday	ARTS-LR6

Membranes


Extracellular fluid
Glycoprotein
Glycolipid
Carbohydrate
Cholesterol
Peripheral protein
Integral protein
Filaments of cytoskeleton
Cytoplasm

Membranes

- The plasma membrane is complex and diverse and has many functions

- The cell goes to some effort and energy expenditure to control the lipid and protein compositions and also the biophysical properties and shape of the membrane. Why?

- Because interactions between proteins and lipids at the plasma membrane must be regulated (just like all interactions in the cell).

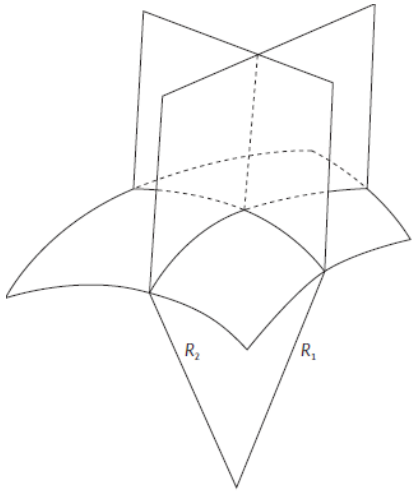
- Because the diverse functions of membranes in the cells (the plasma membrane, Golgi, ER, vesicles) require control of membrane shape and dynamics

- We will look at the mechanics of membranes: curvature and bending as well as the organisation within membranes

- The most basic question is how is membrane curvature defined and controlled

Membrane Curvature

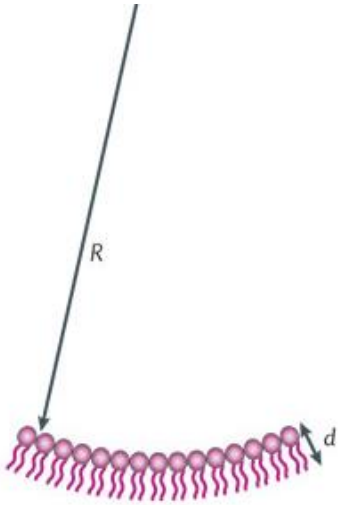
The membrane is a 2-dimensional surface and therefore is described by two perpendicular **Principal curvatures**, c1 and c2

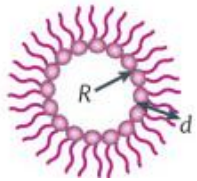


c1 = 1/R1 and c2 = 1/R2

The larger the curvature the smaller the **radius of curvature**

Low curvature





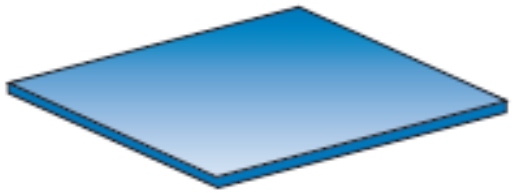
High curvature

Membrane Curvature

The two principal curvatures can be combined to generate two new quantities:

The **Total Curvature**, J = c1 + c2 and the **Gaussian Curvature**, K = c1c2

Obviously if c1 = c2 = 0 then J = 0 and K = 0 and we have a flat membrane



Plane	c_1	c_2	J	K
	0	0	0	0

Membrane Curvature

The two principal curvatures can be combined to generate two new quantities:

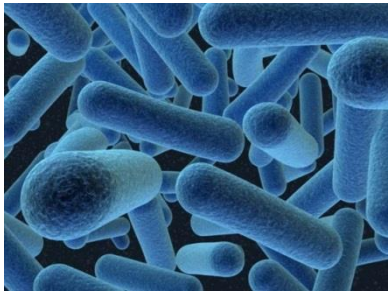
The **Total Curvature**, J = c1 + c2 and the **Gaussian Curvature**, K = c1c2

If the curvature is only in one direction, we produce a cylinder

Cylinder



c_1	c_2	J	K
+	0	+	0



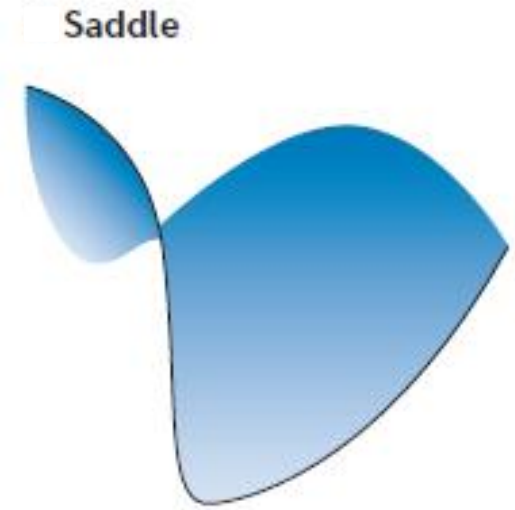


Membrane Curvature

The two principal curvatures can be combined to generate two new quantities:

The **Total Curvature**, J = c1 + c2 and the **Gaussian Curvature**, K = c1c2

Positive and negative curvature produce a saddle:



c_1	c_2	J	K
+	−	0	−



Membrane Curvature

The two principal curvatures can be combined to generate two new quantities:

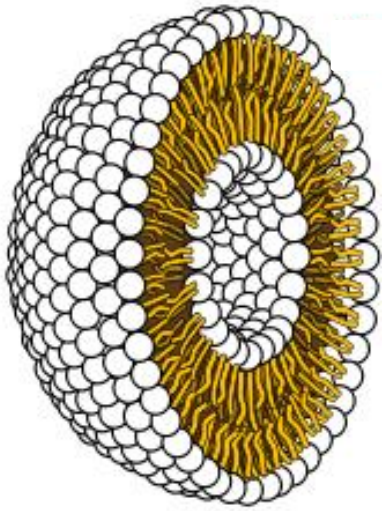
The **Total Curvature**, J = c1 + c2 and the **Gaussian Curvature**, K = c1c2

Two directions of positive curvature produce a sphere:

Sphere



c_1	c_2	J	K
+	+	+	+



Membrane Curvature

It is possible to calculate how much energy is required to transition from one state to another. For example, to go from a planar membrane to a cylindrical structure with radius R and length L, you need:

$$E = \pi L \kappa \left(\frac{1}{R} - 2J_S\right)$$

Here, J_S is known as the spontaneous curvature of the membrane and depends on the lipid types.

κ is the bending modulus of the membrane. It also depends on the lipid types and is especially dependent on the phase with liquid ordered phase bilayers being stiffer.

Importantly, κ is highly dependent on the bilayer thickness (and hence on phase) and scales with d^3.

The value of κ for a typical DOPC disordered bilayer is roughly 5×10^{-20}J

Membrane Curvature

It is possible to calculate how much energy is required to transition from one state to another. For example, to go from a planar membrane to a cylindrical structure with radius R and length L, you need:

$$\mathrm{E} = \pi L \kappa \left(\frac{1}{R} - 2J_S\right)$$

For a hypothetical cylindrical membrane to form spontaneously, E = 0, in which case

$$J_S = \frac{1}{2R}$$

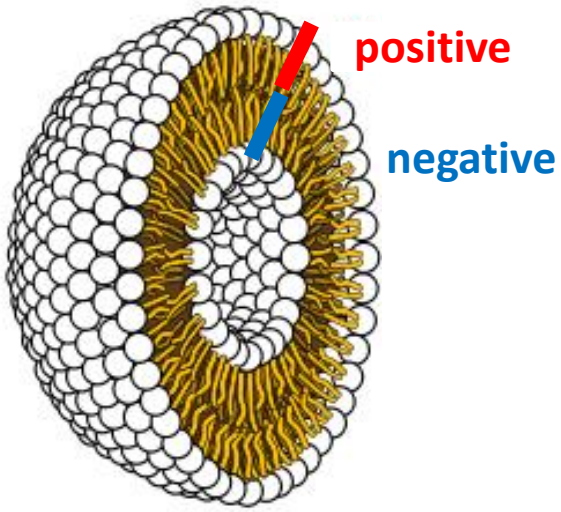
So obviously making large cylinders or spheres requires less energy and hence less spontaneous curvature than making small ones.

But what exactly is J_S for a bilayer?

Membrane Curvature

J_S has two components – one from each component of the bilayer

$$J_S = \frac{1}{2}\left(J^{out} - J^{in}\right)$$



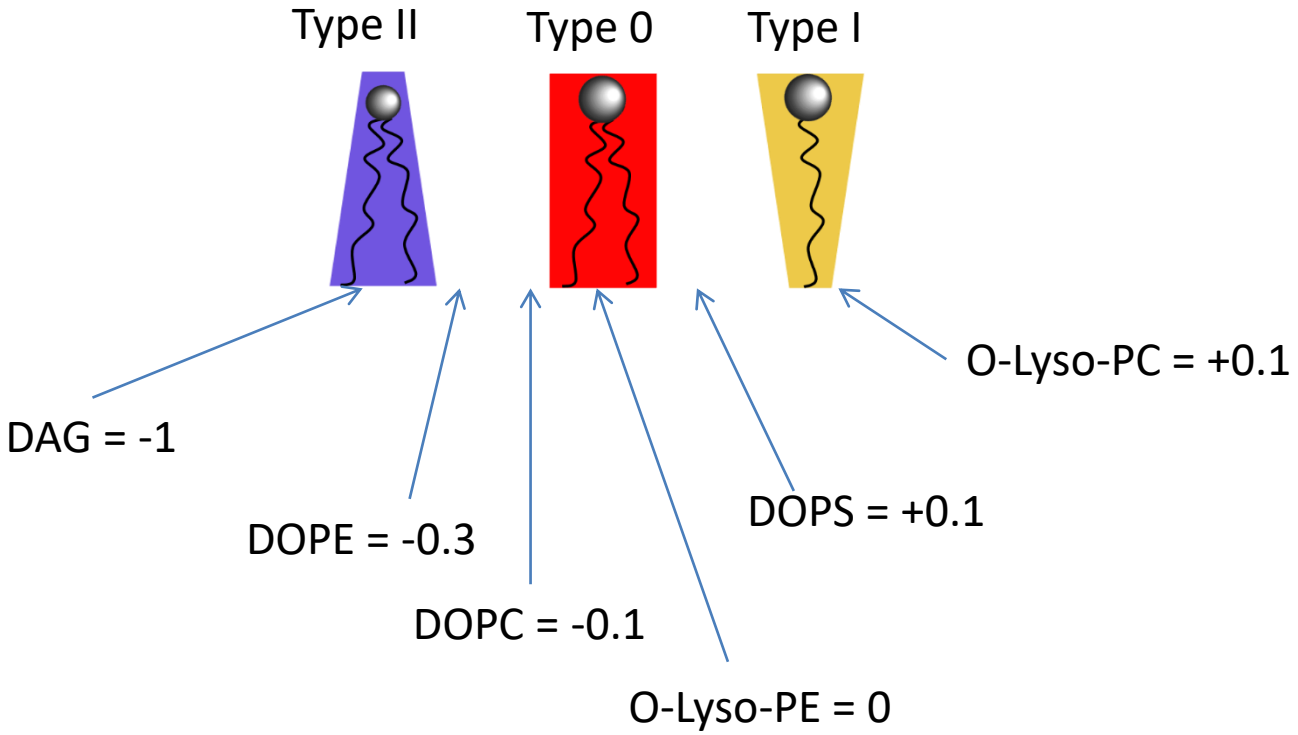
So J_S is basically just the average spontaneous curvature of the two leaflets, corrected for sign

J^{out} and J^{in} depend on the lipid type:

Membrane Curvature

The most basic mechanism to control membrane curvature is to exploit the spontaneous curvature of membranes containing different types of lipids

So each type of lipid has a different spontaneous curvature, J, and therefore will favour different kinds of structures



Membrane Curvature

As in the case for a cylinder, there is an expression for the energy required to transform a planar bilayer into a spherical one

$$E = 8\pi\left(\kappa + \frac{1}{2}\kappa_G\right) - 8\pi\kappa R J_S$$

Sphere



c_1	c_2	J	K
+	+	+	+

κ_G comes from the fact we have Gaussian curvature and is called the elastic modulus of Gaussian curvature. However, it is basically impossible to measure experimentally

For spontaneous formation of this sphere, we therefore need

$$J_S = \frac{1}{R}\left(1 + \frac{\kappa_G}{2\kappa}\right)$$

Some very estimated values puts κ_G around -6 x 10^{-20} J.

Membrane Curvature

We have seen how membrane curvature can be defined and how lipid spontaneous curvature can allow the formation of planar bilayers, cylinders and spheres.

For general structural components this is all fine however, it is not very useful for the cell for the generation of dynamic processes

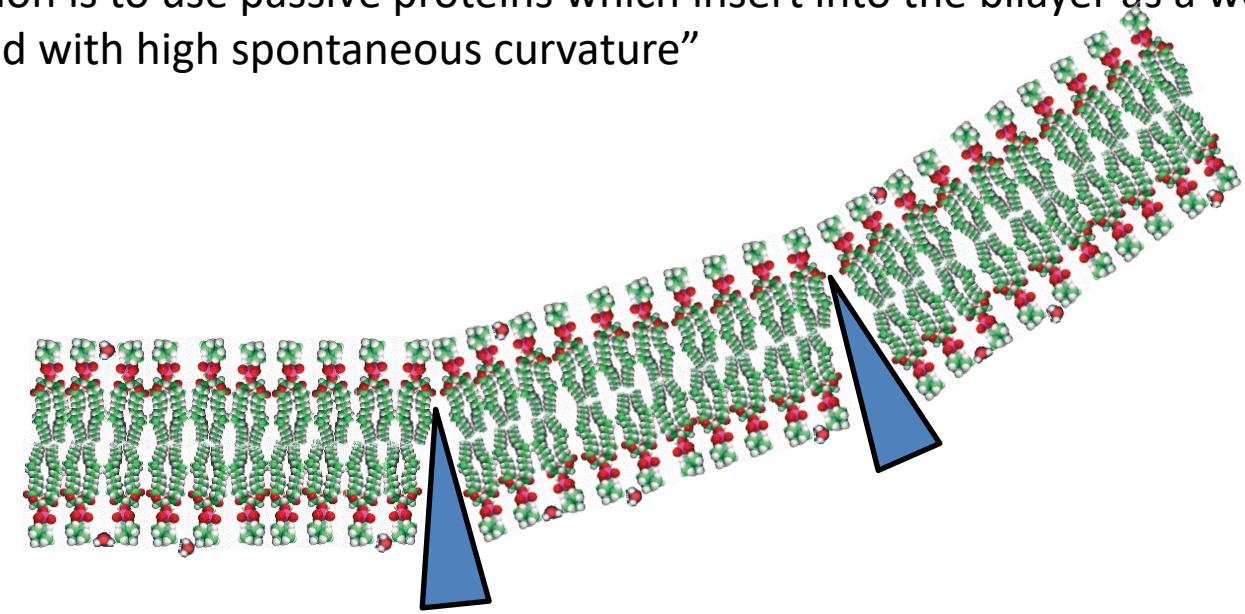
For example, if the cell wants to pinch off a small vesicle from the plasma membrane or Golgi or to remodel the plasma membrane during cell migration

This is because the cell would have to change the lipid composition of the bilayer. It would have to undergo a complex series of regulated reactions to synthesis the required lipids and traffic them to the correct membranes. This would be slow and inefficient

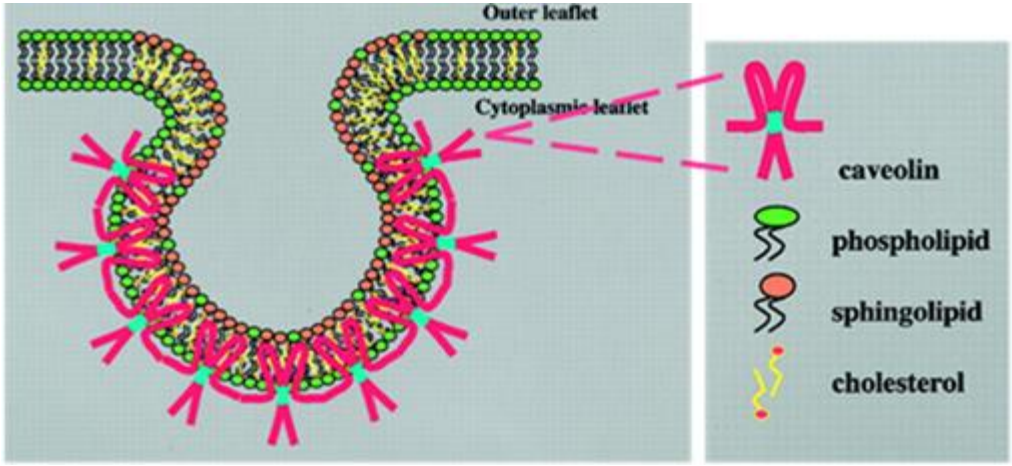
To dynamically regulate membrane curvature, the cell uses proteins

Membrane Curvature

One option is to use passive proteins which insert into the bilayer as a wedge, essentially acting as a “lipid with high spontaneous curvature”



Some good examples are the Caveolin proteins and the Flotillins



Membrane Curvature

The proteins change the spontaneous curvature of the bilayer and therefore can be treated in the same was as lipids

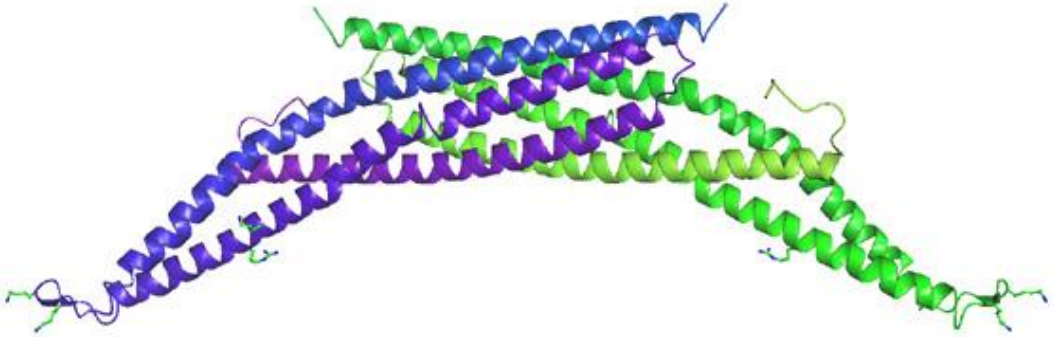
They can insert into either leaflet of the membrane to control the direction of curvature

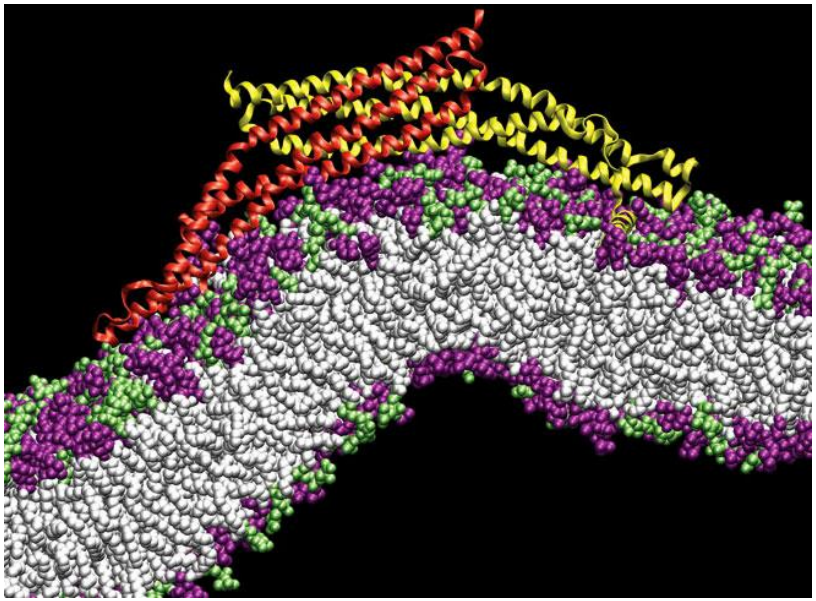
However, unlike lipids, they can be selectively regulated so that membrane curvature can be rapidly changed

A second strategy is the use of scaffold proteins. These do not insert into the membrane

There are many examples of these proteins, but many share some common, evolutionarily conserved domains e.g. the BAR domain (Bin-Amphiphysin-R)

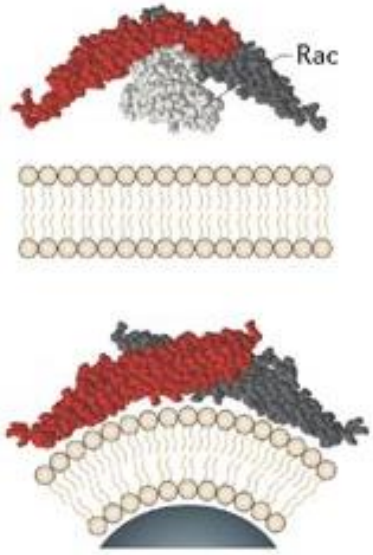
Membrane Curvature



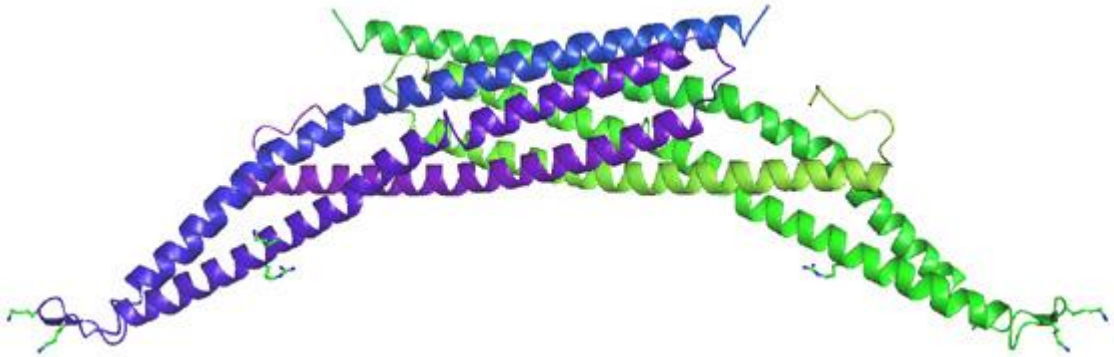


The concave surface has high affinity for phospholipid head groups e.g. negatively charged phosphatidylserine

BAR domains can also be regulated to selectively bend membranes e.g. the binding of Rac proteins



Membrane Curvature



In addition to their role in bending membranes, BAR domains can have an additional role in targeting proteins to membranes with specific curvature. I.e. They can act as curvature sensors

Some types of BAR domain are flexible and can alter their curvature

BAR domain containing proteins often have other domains which confer additional abilities e.g. binding to the actin cytoskeleton or to motor proteins

Endo and exocytosis

Some of the most important uses of selectively bending membranes is in **endocytosis** and **exocytosis**

Endo and exocytosis

Some of the most important uses of selectively bending membranes is in **endocytosis** and **exocytosis**

Endocytosis is a method of transporting substances into the cell

There are many different forms, but probably the most common is the use of caveolae

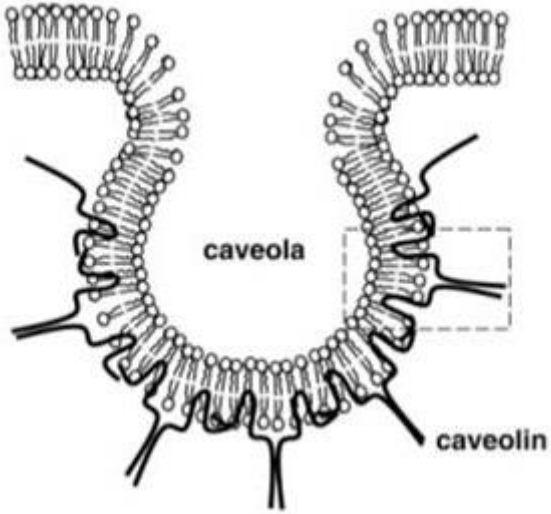
Caveolae are small (~50nm) flask shaped invaginations on the cell surface



Endo and exocytosis

The membranes in Caveolae are highly ordered and enriched in Cholesterol – they are therefore usually considered a special kind of lipid raft

The curvature is maintained by the 21kD protein Caveolin (or which there are 3 isoforms) using the wedge-insertion method



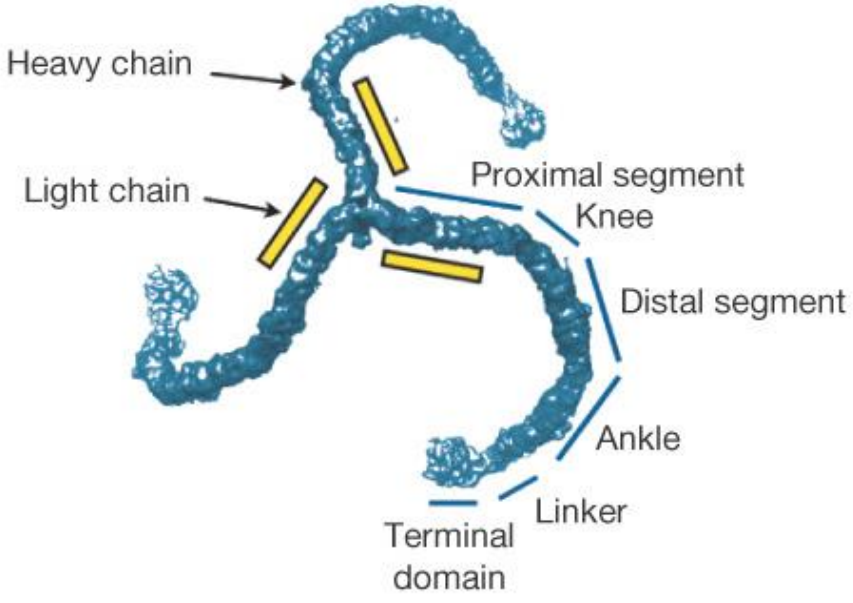
The neck of the flask can then be pinched off to form an intracellular vesicle

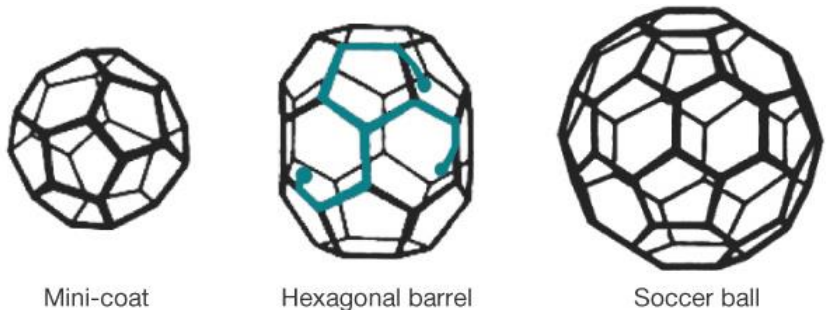
Endo and exocytosis

Caveolae are just one mechanism of endocytosis

Another common mechanism is called **clathrin-mediated endocytosis**.

The major use of clathrin-mediated endocytosis is to uptake molecules which have bound to specific receptors on the cell surface. Hormone receptors are a good example. Hence it is sometimes called **receptor mediated endocytosis**.





Endo and exocytosis

